$e \quad a \quad a^2 \quad b \quad b^2$

$ab \neq e, a^2, b, b^2.$

ab

$ab^2 \neq e, a^2, b, b^2, ab$

ab^2

if $ba = ab$, $\langle a, b \rangle < G$ is abelian, but this implies that $\langle a, b \rangle = G$ is abelian. So:

$ba \neq e, ab, a, a^2, b, b^2.$

Can $ba = ab^2$? If so, then: $ababab = e$ ← All elements of order 3

$ababa\cancel{b} = b^2$

$$\left[\begin{array}{c} \text{if } ba = ab^2 \Rightarrow \\ ababa = a\,ab^2\,ab^2. \end{array}\right]$$

$\rightarrow a^2 b^2 a = e.$

$\rightarrow a^2 b^2 = a^2.$ ↯.

$\rightarrow ba \neq ab^2.$

ba

Now ba^2.

We know $ba^2 \neq e, a, a^2, b, b^2, ba.$

Can $ba^2 = ab$? If so:

$e = abab\,ab = ba^2abab = b^2ab$; $e = b^2ab$; $b^2 = b^2a$ ↯

Can $ba^2 = ab^2$?

$a^2 = ab^2ba = ba^2ba$; $a^2 = ba^2ba$; $e = ba^2ba^2 \Rightarrow$

order of ba^2 would be 2 ↯.

$\Rightarrow ba^2$ is a new element.

Now one creates the table:

$\cdot$	e	a	a^2	b	b^2	ab	ab^2	ba	ba^2	
e	e	/	/	/	/	/	/	/	/	
a	/	a^2	e	ab	ab^2	a^2b	a^2b^2	aba	aba^2	
a^2	/	e	a	a^2b	a^2b^2	b	b^2	a^2ba	a^2ba^2	
b	/	ba	ba^2	b^2	e	bab	bab^2	b^2a	b^2a^2	
b^2	/	b^2a	b^2a^2	e	b	b^2ab	b^2ab^2	a	a^2	
ab	/	aba	aba^2	ab^2	a	$abab$	$abab^2$	ab^2a	ab^2a^2	
ab^2	/	ab^2a	ab^2a^2	a	ab	ab^2ab	ab^2ab^2	a^2	e	
ba	/	ba^2	b	bab	bab^2	ba^2b	ba^2b^2	$baba$	$baba^2$	
ba^2	/	b	ba	ba^2b	ba^2b^2	b^2	e	ba^2ba	ba^2ba^2	

Now one starts with considerations:

- $ab^2ab^2 = (ab^2)^{-1} = ba^2$.
- $abab = (ab)^{-1} = b^2a^2$.
- Who can be $(ab)^{-1}$?

$\cancel{e}$ $\cancel{a}$ $\cancel{a^2}$ $\cancel{b}$ $\cancel{b^2}$ $\cancel{ab}$ $\underbrace{ab^2}$ $\underbrace{ba}$ $\cancel{ba^2}$

$abab^2 \overset{?}{=} e \Rightarrow aba = b \Rightarrow ba = a^2b$.

But $baa^2b = b^2 \Rightarrow baba = b^2$ $b^2aba = e$ $aba = e$

But inverse of a is a^2, not ab. So $(ab)^{-1} = ba$.

et cetera...